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SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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AI-Powered Platform for Online Food Delivery Management

CSA1501:CLOUD COMPUTIG

Guided By;
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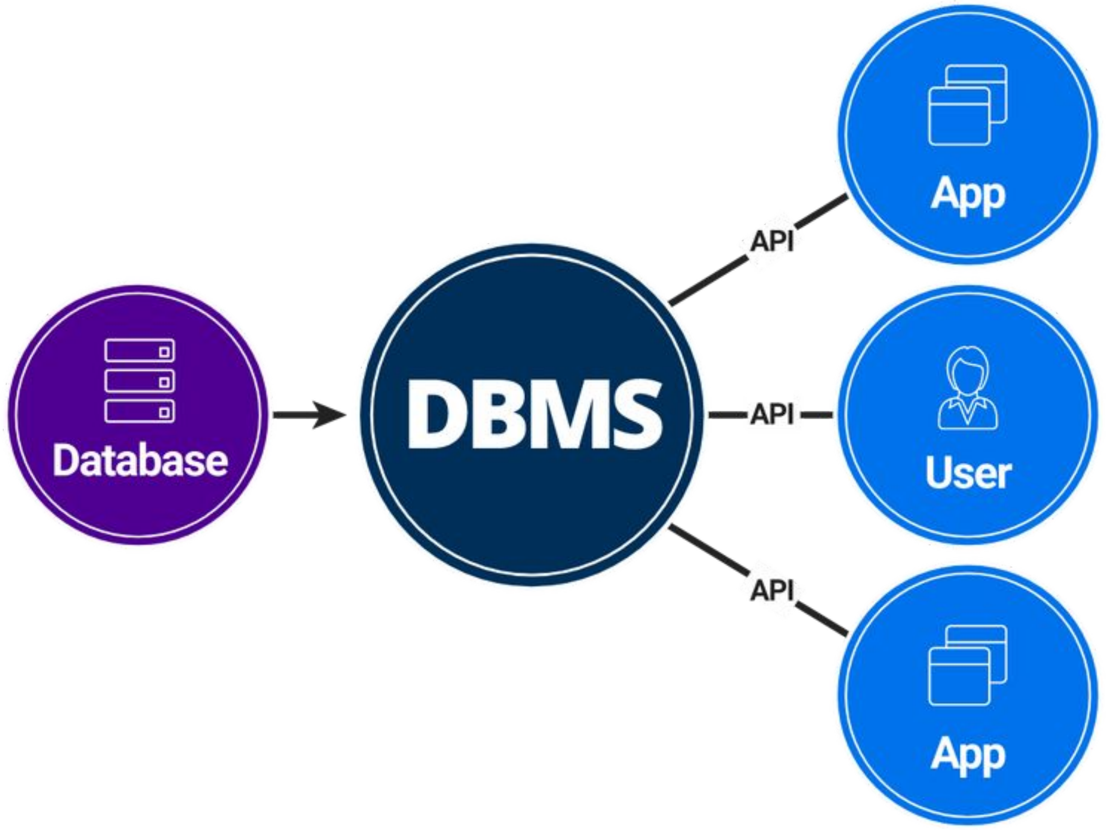
SIMATS Engineering



Presented by
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Aim and Objective

AIM: To design and implement a full stack AI powered Food Delivery Management System that demonstrates core DBMS principles including normalized schema design, ACID transactions, role-based access control, and intelligent recommendation systems within a real-world application.

OBJECTIVES:

- Design a normalized relational database schema (3NF) covering users, restaurants, orders, payments, and delivery systems.
- Implement referential integrity using Primary Keys, Foreign Keys, and constraints.
- Develop a real time order processing system with transaction safety.
- Implement an Indian billing system including GST, delivery charges, and platform fees.
- Provide a role based system for Customer, Admin, Restaurant, and Delivery Partner.

Problem Statement

Traditional food delivery systems face several problems such as:

- lack of personalized recommendations,
- inefficient order management,
- redundant database storage,
- security risks,
- and transaction failures.

"Our project aims to solve these problems using an efficient DBMS design."

Proposed Solution

“Our solution uses a normalized relational database with MySQL”.

We implemented:

- Primary and Foreign Keys for integrity,
- GPS-based restaurant filtering,
- Indian billing system,
- JWT-based authentication,
- and ACID-compliant transaction handling.

These features improve security, consistency, and user experience.”

Introduction

- Food delivery systems have become an important part of modern life by providing fast and convenient access to food services.
- Traditional food delivery platforms often face problems such as delayed deliveries, poor order management, and inefficient restaurant handling.
- Our project introduces an AI powered Online Food Delivery Management System to improve speed, accuracy, and customer satisfaction.
- The system allows users to register, browse restaurants based on location, place orders, manage carts, and make secure payments.
- Restaurants can efficiently manage menus, update food availability, and handle customer orders through the platform.
- The project uses technologies such as MySQL LITE , HTML, CSS, JavaScript, Python Flask/Node.js, and RESTful APIs for development.

FOODOO™

AI-POWERED INDIAN FLAVORS

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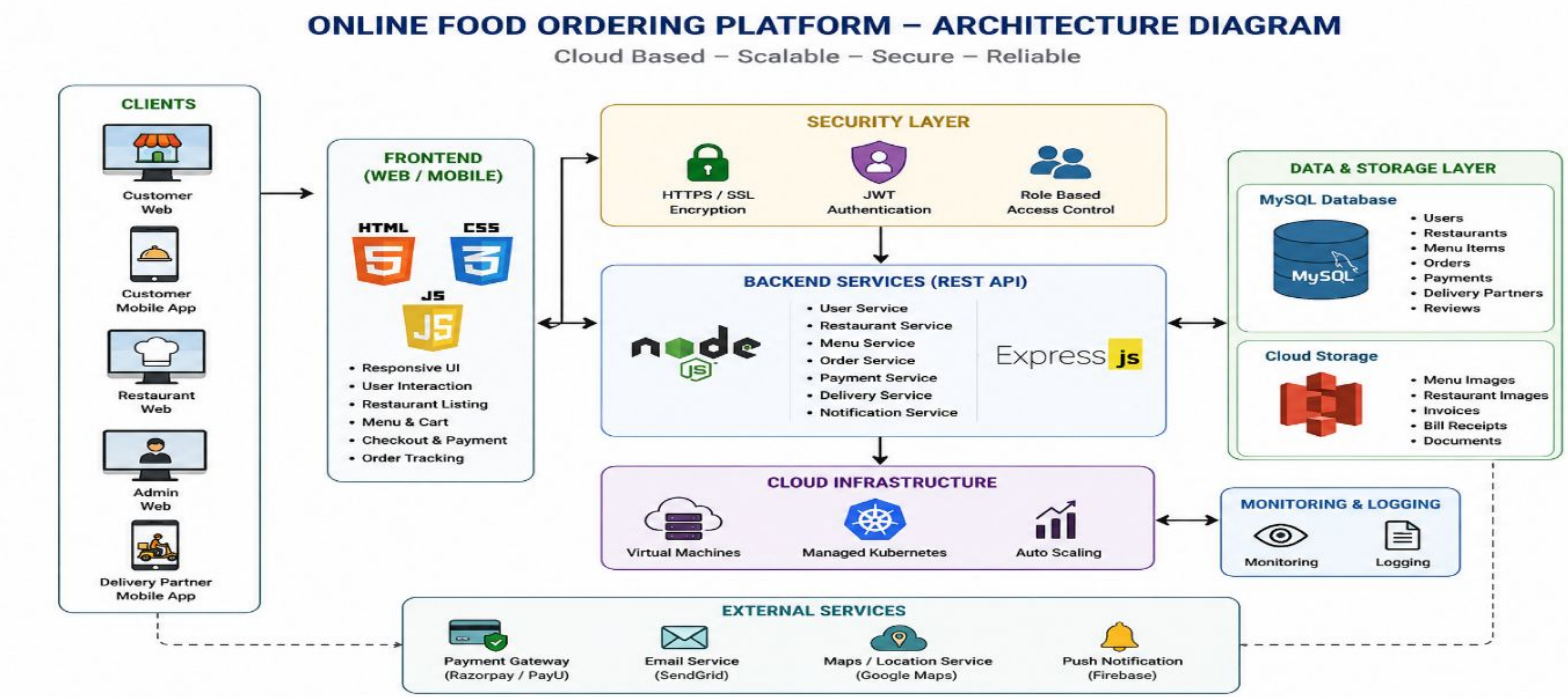
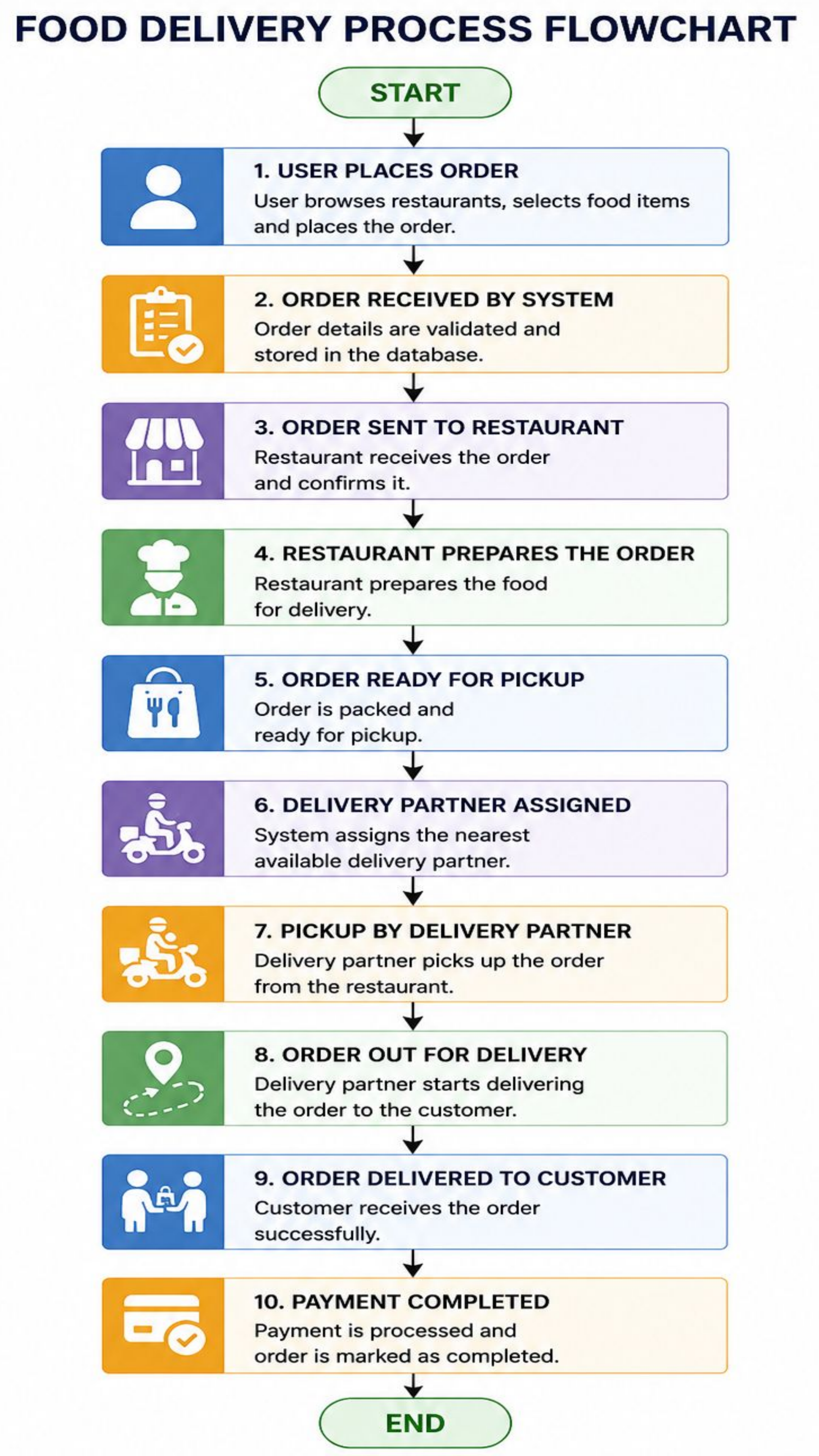
System Architecture



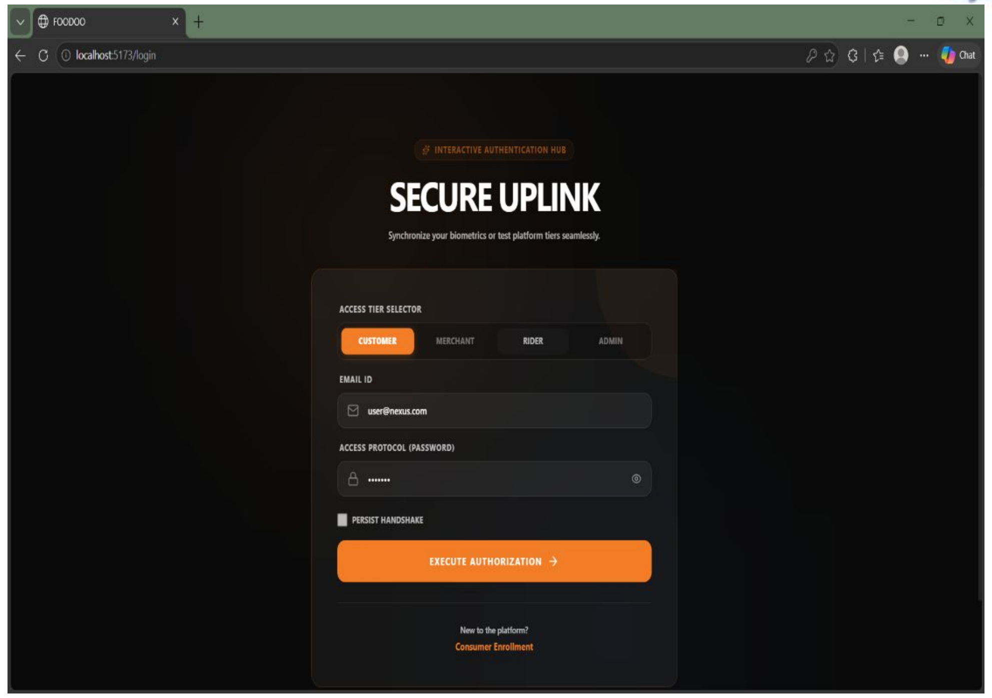
Module 1
User & Restaurant Management Module

Overview of the User & Restaurant Management Module

- This module manages customer registration, login, and profile handling.
- Restaurants can register, upload menus, update food availability, and manage orders.
- The module stores and retrieves restaurant details based on location filtering.
- Input: User details, restaurant details, menu information.
- Output: Registered users, restaurant listings, updated menu database.
- Role: Acts as the foundation of the complete food delivery platform.
- Ensures secure authentication and efficient database management for smooth system operation.

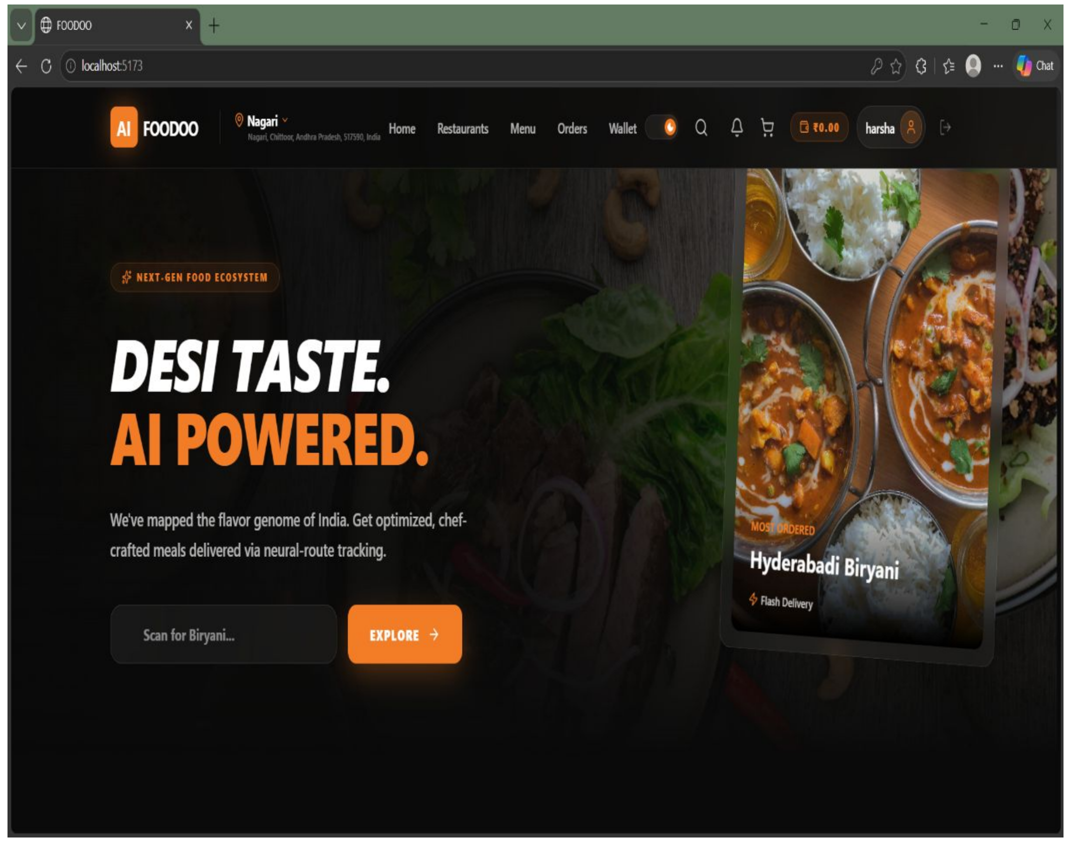


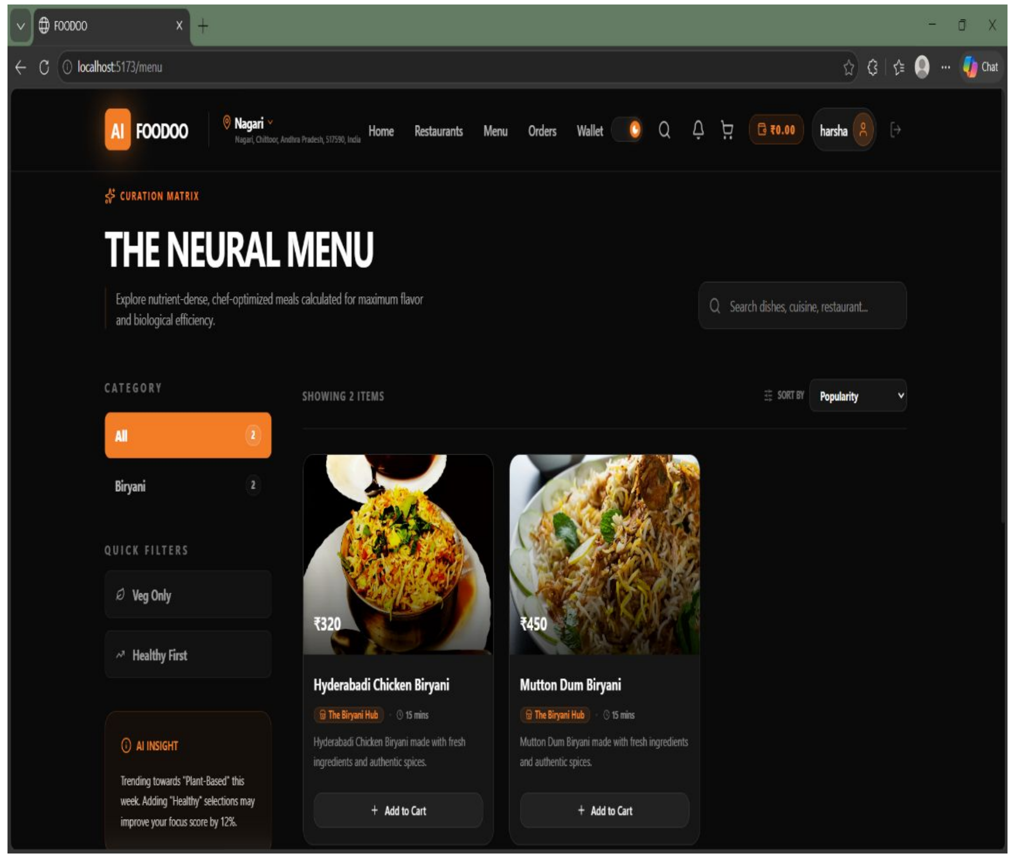
Implementation & Output

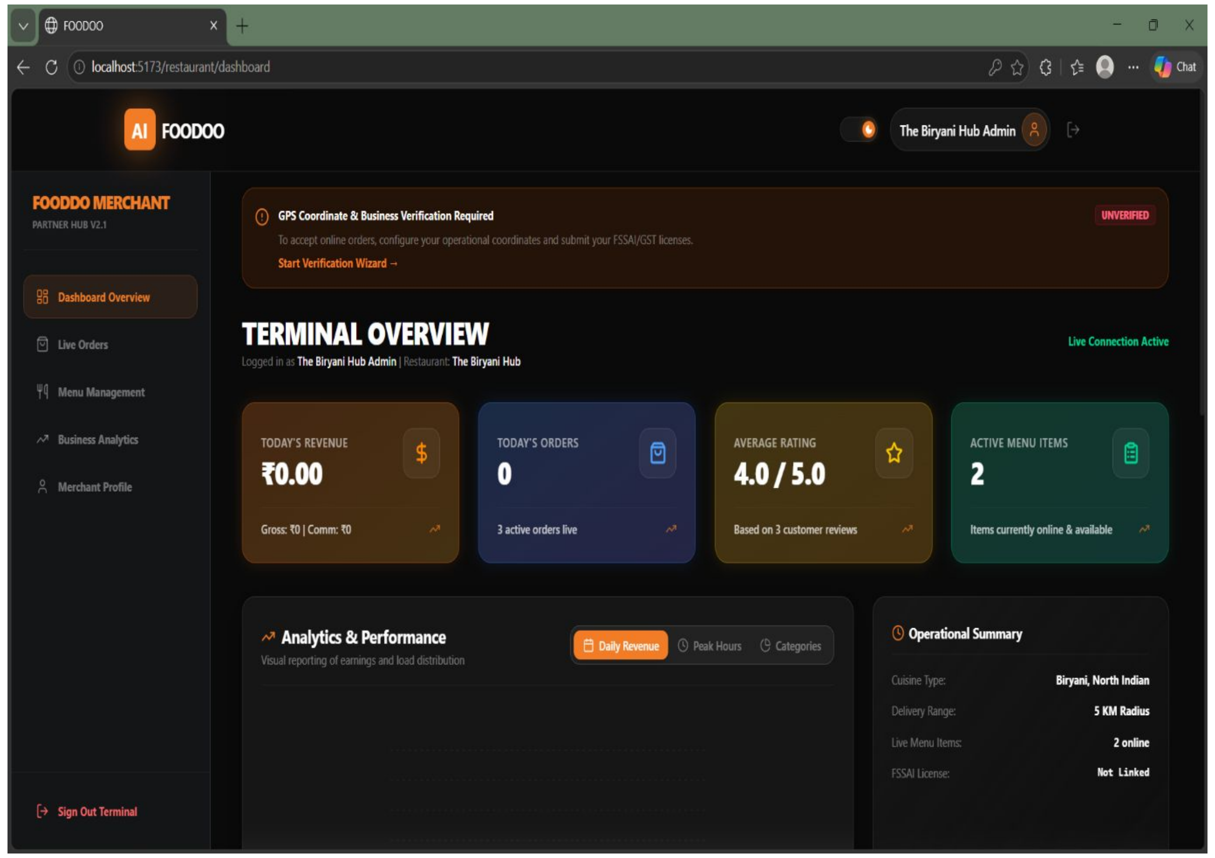


Implementation Details

- Implemented user authentication and session management.
- Created restaurant dashboard for menu management.
- Developed location based restaurant filtering.
- Integrated MySQL database for storing user and restaurant data.





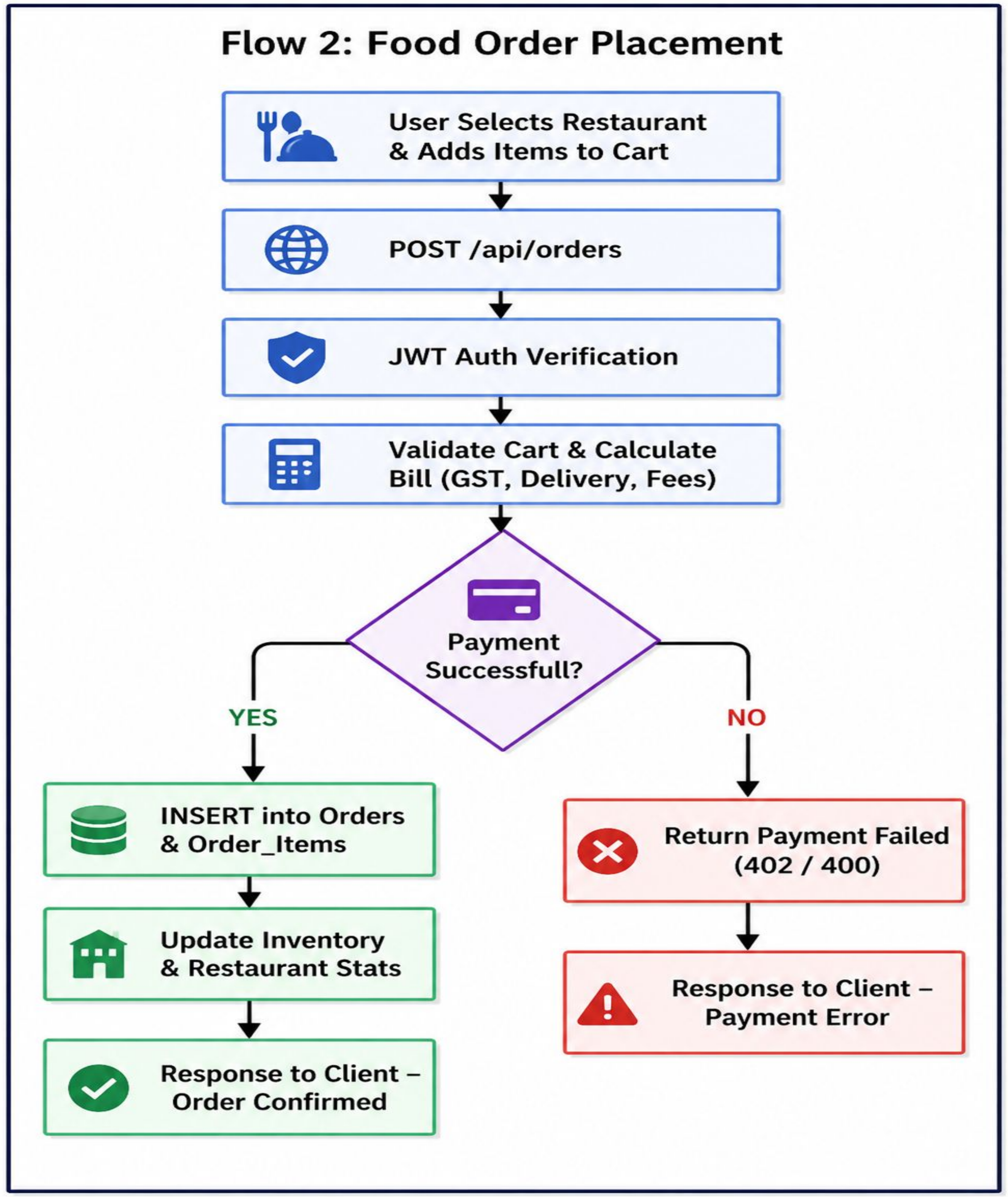


Module 2

Order Processing & Billing Module

Overview of Order Processing & Billing Module

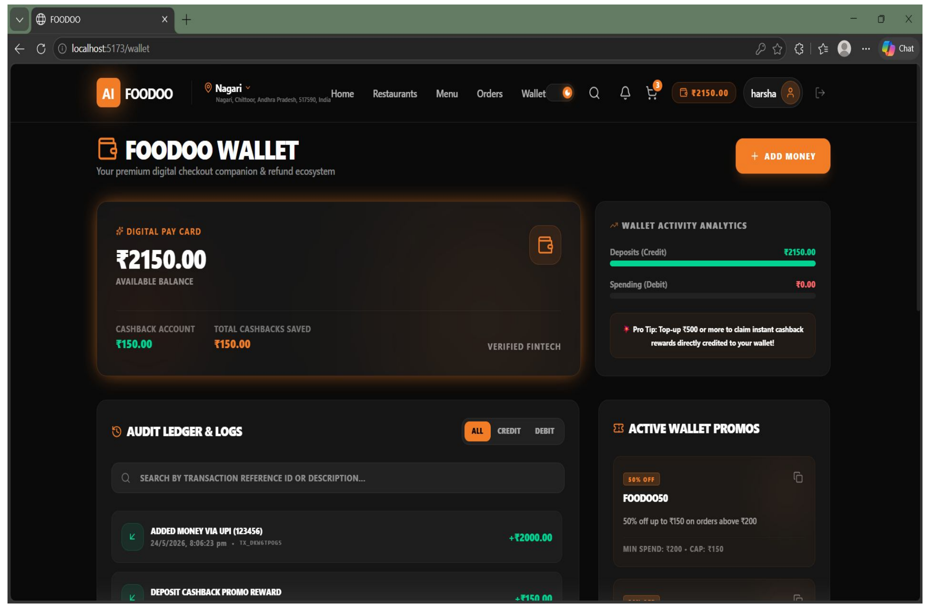
- Handles the complete lifecycle of an order from cart management to final delivery tracking.
- Uses data from Module 1 including user details, restaurant information, and menu items.
- Allows users to add items to cart, modify quantities, and place orders efficiently.
- Output includes confirmed orders, billing information, and updated order status.
- Ensures smooth and efficient transaction processing within the system.

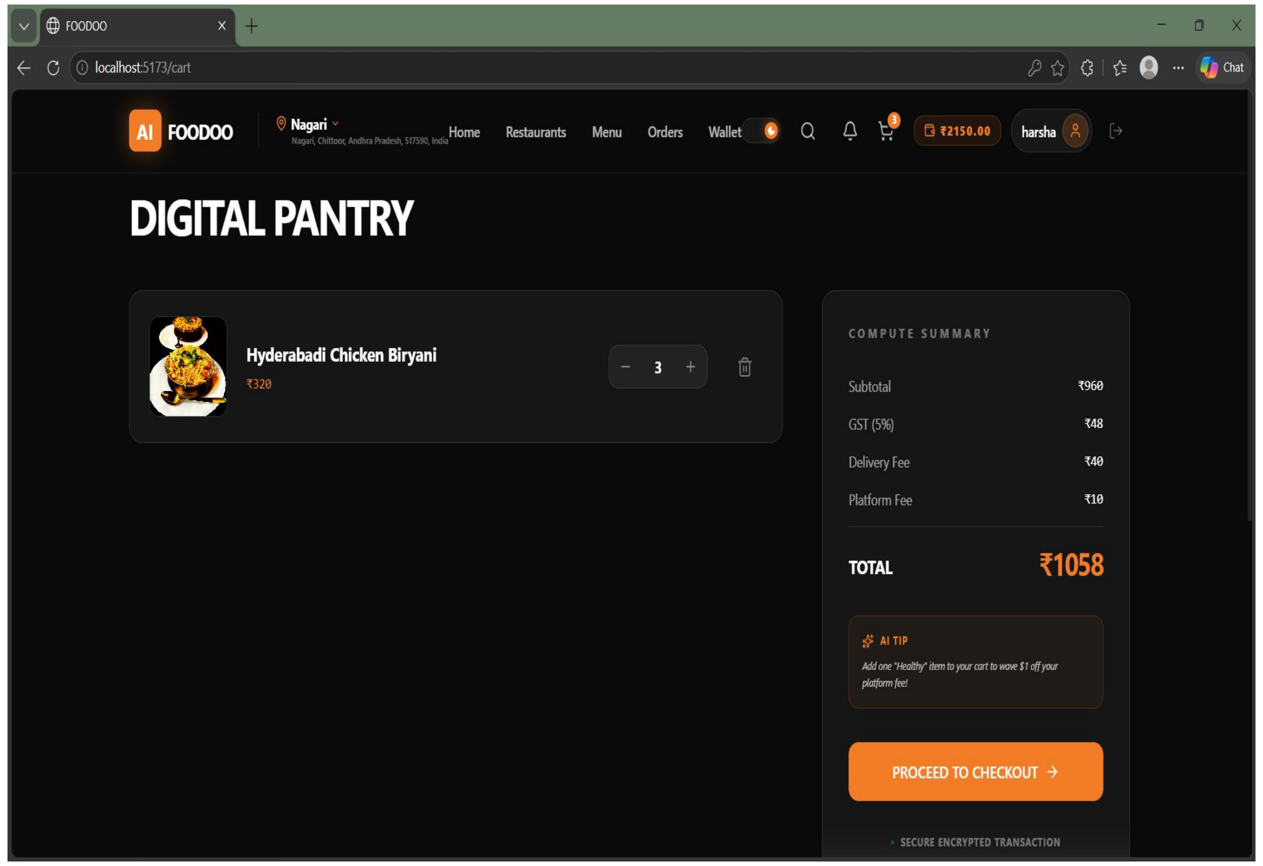


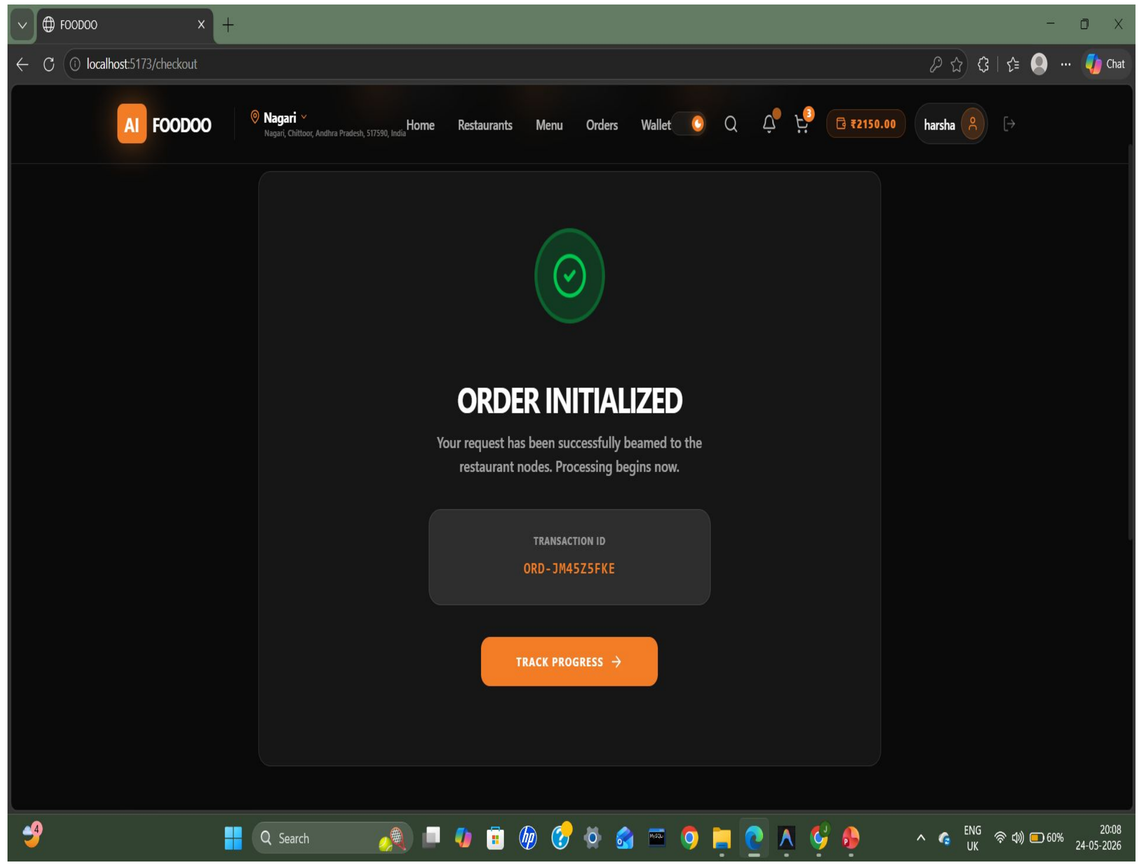
Implementation & Output

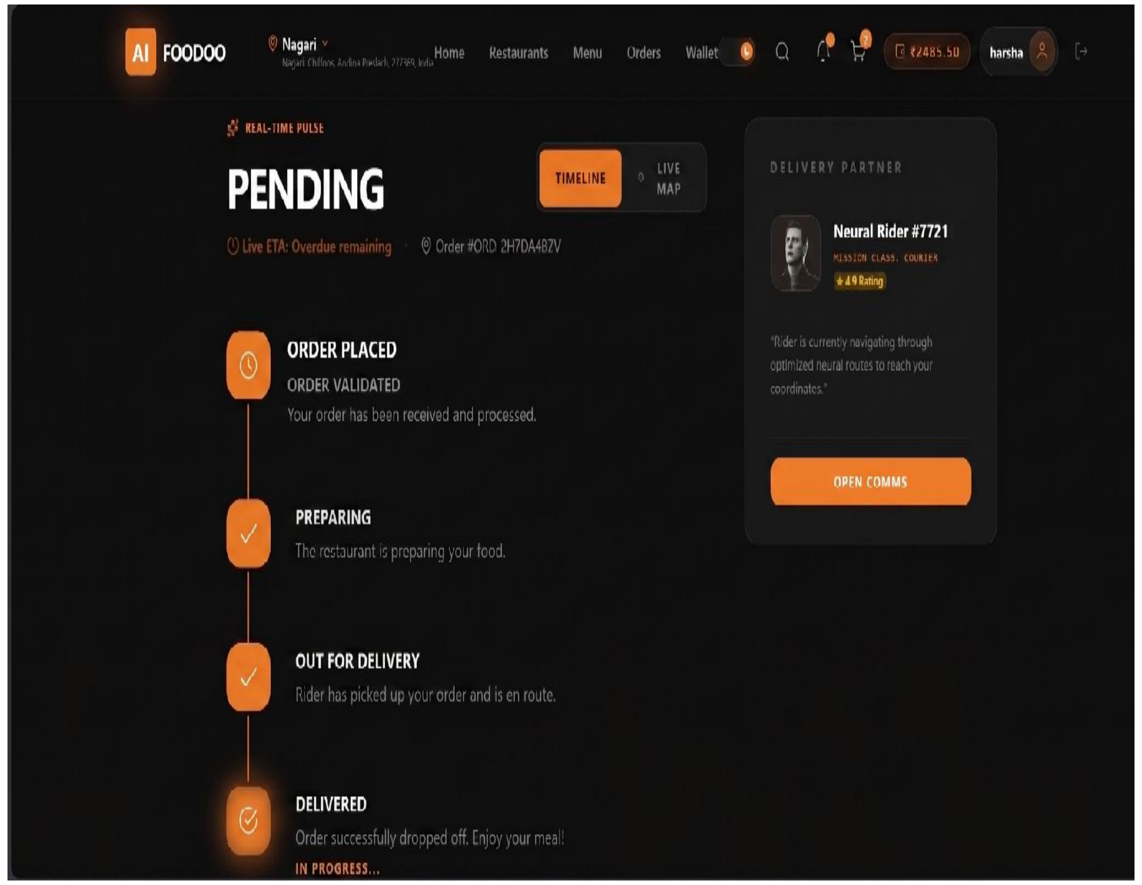
Final Output

- Complete food ordering and delivery workflow implemented successfully.
- Real time order tracking and payment status updates achieved.
- AI-based recommendations improved customer engagement.
- Delivery management reduced processing delays.



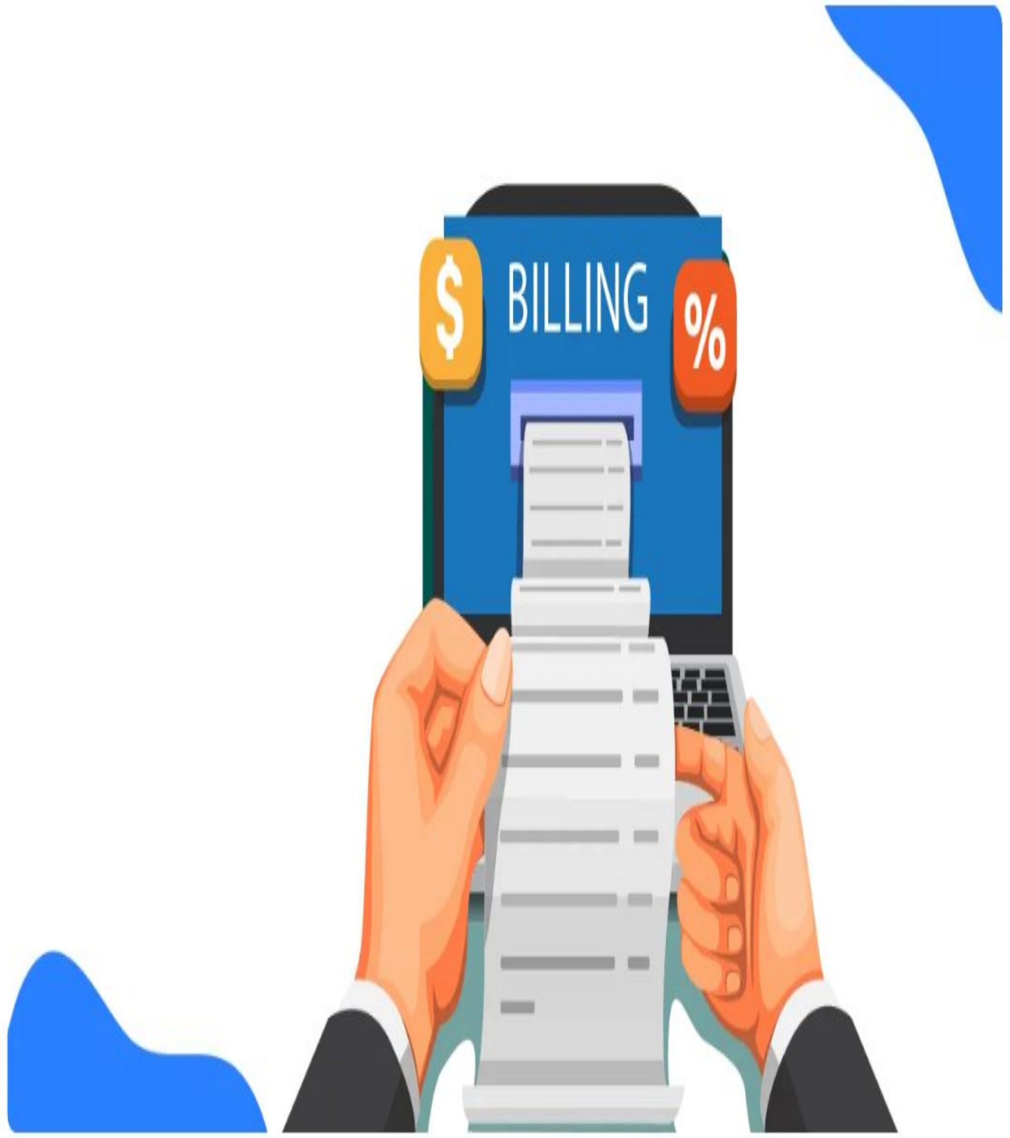






Key Modules & Features

- **Restaurant Management:** Enables restaurant owners to register, manage menus, update item availability, and handle incoming orders efficiently.
- **Order Processing:** Allows users to add items to cart, place orders, and track order status from preparation to delivery.
- **Billing System:** Implements a structured Indian billing system including GST (5%), delivery charges, platform fees, and packaging costs for transparent pricing.
- **Payment Handling:** Manages order payments with transaction safety and maintains payment history for each user.
- **Location Based Services:** Uses GPS based location detection to display nearby restaurants and optimize delivery operations.
- **AI Recommendation System:** Suggests food items based on user preferences, order history, and trending data to enhance user experience.



Advantages

- Uses a normalized database (3NF) to ensure minimal redundancy and consistent data storage.
- Provides AI based food recommendations based on user preferences and behavior.
- Implements a transparent billing system with GST, delivery charges, and platform fees.
- Enables location based services to find nearby restaurants using GPS.

Limitations

- Uses simulated payment systems instead of real-time payment gateway integration.
- The AI recommendation model is basic and may not provide highly accurate results.
- System scalability may be limited without cloud infrastructure support.

Future Enhancements

- **Live Payment Gateway Integration** Integrate real payment systems such as Razorpay or Stripe for secure online transactions, refunds, and payment tracking.
- **Mobile Application Development** Develop cross platform mobile apps (Android/IOS) using React Native or Flutter for better accessibility and user experience.
- **Real Time GPS Tracking** Implement live order tracking using maps to allow users to monitor delivery status in real time.
- **Advanced AI Recommendation System** Enhance the recommendation engine using machine learning models to provide more accurate and personalized suggestions.
- **Cloud Deployment & Scalability** Deploy the system on cloud platforms (AWS, Azure, or GCP) for improved scalability, performance, and availability.
- **Notification System Integration** Implement email/SMS/push notifications for order updates, offers, and delivery alerts to improve user engagement.

Conclusion

- Foodoo successfully demonstrates the practical implementation of core relational database principles in a real world food delivery management system.
- The project integrates normalized schema design, ACID transactions
- The system provides a complete end to end solution, including user management, restaurant operations, order processing, billing, and delivery tracking.
- A structured Indian billing system ensures transparency and accuracy in financial transactions with proper tax and fee breakdown.
- AI based recommendation features and location based services enhance user experience and improve operational efficiency.
- Overall, this makes effectively bridges theoretical DBMS concepts with modern full stack development, delivering a scalable and user friendly solution.

References

- Li, C., Mirosa, M., & Bremer, P. (2023). Review of online food delivery platforms and their impacts on sustainability. *Sustainability*, *12*(14), 5528.
- Shankar, A., Jebarajakirthy, C., Nayal, P., Maseeh, H. I., Kumar, A., & Sivapalan, A. (2022). Online food delivery: a systematic synthesis of literature and a framework development. *International Journal of Hospitality Management*, *104*, 103240.
- Chai, L. T., & Yat, D. N. C. (2023). Online food delivery services: Making food delivery the new normal. *Journal of Marketing advances and Practices*, *1*(1), 62-77.
- Shroff, A., Shah, B. J., & Gajjar, H. (2022). Online food delivery research: a systematic literature review. *International Journal of Contemporary Hospitality Management*, *34*(8), 2852-2883.
- Yeo, V. C. S., Goh, S. K., & Rezaei, S. (2022). Consumer experiences, attitude and behavioral intention toward online food delivery (OFD) services. *Journal of Retailing and Consumer services*, *35*, 150-162.
- Gavilan, D., Balderas-Cejudo, A., Fernández-Lores, S., & Martinez-Navarro, G. (2024). Innovation in online food delivery: Learnings from COVID-19. *International journal of gastronomy and food science*, *24*, 100330.

Thank You